HV Port and HV Bridge

File: hv_bridge.kicad_sch

LV Port and LV Bridge

File: lv_bridge.kicad_sch

Gate Drive and Isolated Bias

File: gate_drive.kicad_sch

Sensing and Hardware Protection

File: sensing_protection.kicad_sch

Controller Interface, Aux Power, Comms

File: control_aux.kicad_sch

# SiC 6.6 kW BIDIRECTIONAL DUAL ACTIVE BRIDGE DC–DC CONVERTER – INDUSTRIAL HVDC

HV port 650–850 V DC | LV port 300–450 V DC | 6.6 kW both directions | 100 kHz | SiC 1200 V both bridges

HV bridge 4 x C3M0032120K, LV bridge 4 x C3M0016120K, series inductance 64 uH (4 uH leakage + 60 uH external), n = 2 (E65/3C95, external magnetics module)

Gate drive: 4 x UCC21520 reinforced isolated dual drivers + 8 x isolated +15/–3 V bias; hardware OC/OV comparators –> PROT_DIS –> driver DIS and DSP trip zone

Sensing: AMC1301 isolated amplifiers (V1, V2, I2 shunt), CT on the series inductor (I_L); controller: LAUNCHXL–F280049C BoosterPack headers

Aux: 24 V in –> 12 V (3 A) / 5 V / 3.3 V; CAN (TCAN332), UART service port, precharge contactor outputs and interlock inputs

Sheets: 2 HV port and bridge, 3 LV port and bridge, 4 gate drive and bias, 5 sensing and protection, 6 controller/aux/comms. Generated by dabtools/kicad_dab.

WARNING: HV_* nets up to 850 V DC, LV_* nets up to 450 V DC, gate bias nets ride on the switch nodes. Isolation per 15_Safety_Reliability/01.

6.6 kW, 650–850 V <–> 300–450 V, 100 kHz, SiC DAB, SPS/EPS modulation

Root sheet

**Open hardware – SiC PSFB reference design**

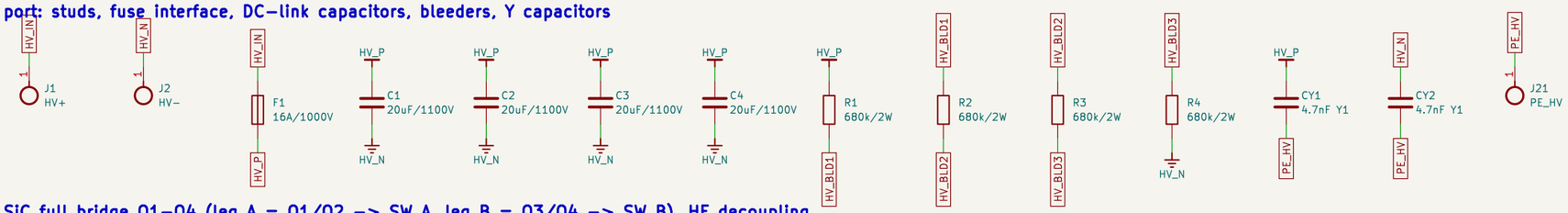
Sheet: /

File: SiC_DAB_6k6.kicad_sch

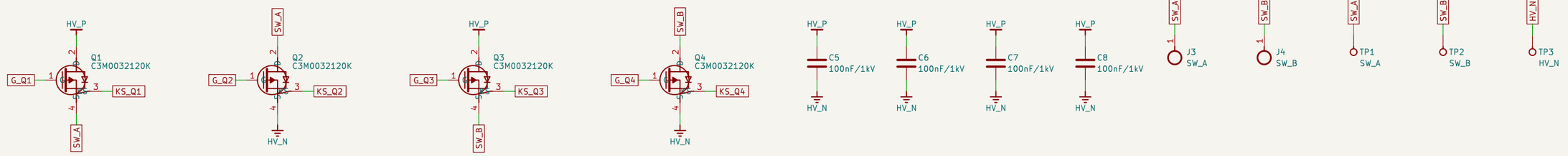
***Title: SiC 6.6 kW Bidirectional Dual Active Bridge, 800 V / 400 V***

| | | |
|---|---|---|
| Size: A3 | Date: 2026–10–09 | **Rev: A** |
| KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4 | | Id: 1/6 |

## HV port: studs, fuse interface, DC-link capacitors, bleeders, Y capacitors




## HV SiC full bridge Q1–Q4 (leg A = Q1/Q2 -> SW_A, leg B = Q3/Q4 -> SW_B), HF decoupling



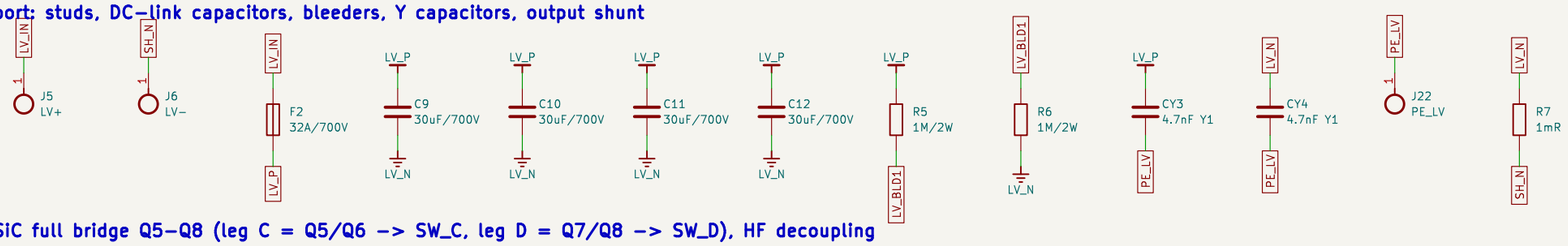

6.6 kW, 650–850 V <-> 300–450 V, 100 kHz, SiC DAB, SPS/EPS modulation
HV Port and HV Bridge
**Open hardware — SiC PSFB reference design**

Sheet: /HV Port and HV Bridge/
File: hv_bridge.kicad_sch

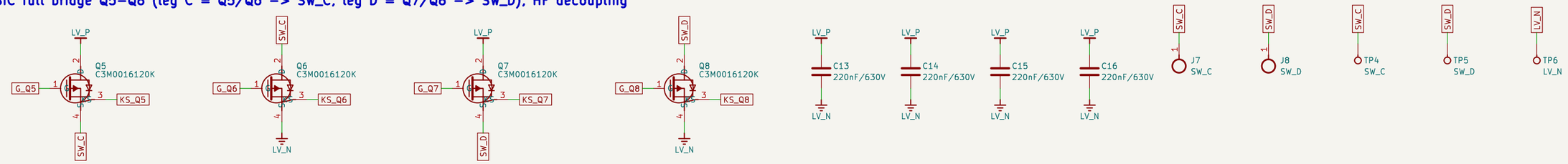
***Title: SiC 6.6 kW Bidirectional Dual Active Bridge, 800 V / 400 V***

| Size: A3 | Date: 2026-10-09 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 2/6 |

## LV port: studs, DC-link capacitors, bleeders, Y capacitors, output shunt




## LV SiC full bridge Q5–Q8 (leg C = Q5/Q6 -> SW_C, leg D = Q7/Q8 -> SW_D), HF decoupling



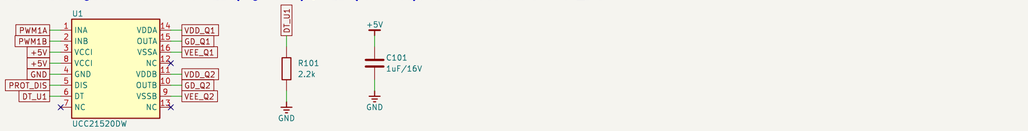

6.6 kW, 650–850 V <-> 300–450 V, 100 kHz, SiC DAB, SPS/EPS modulation
LV Port and LV Bridge
**Open hardware — SiC PSFB reference design**

Sheet: /LV Port and LV Bridge/
File: lv_bridge.kicad_sch

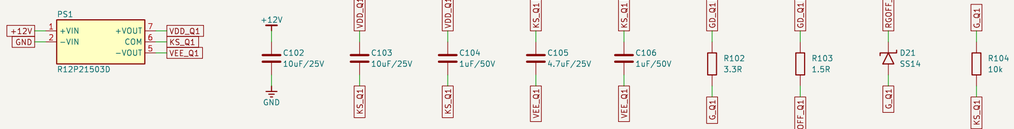
***Title: SiC 6.6 kW Bidirectional Dual Active Bridge, 800 V / 400 V***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-09 | **Rev: A** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 3/6 |

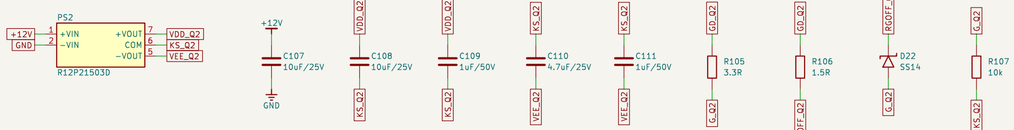
## Isolated dual gate driver U1 for Q1 (high side) / Q2 (low side), fault disable via PROT_DIS



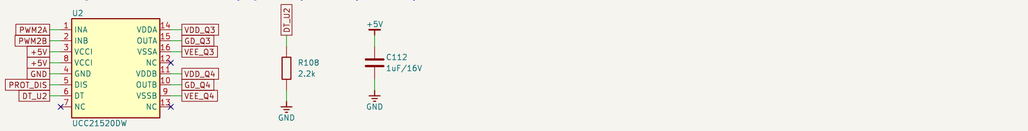

## Q1 isolated bias (+15 V / −3 V) and gate network



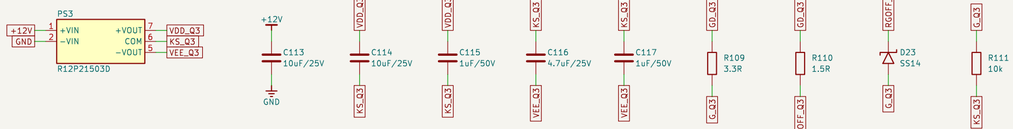

## Q2 isolated bias (+15 V / −3 V) and gate network



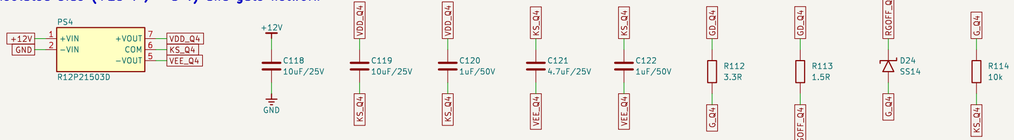

## Isolated dual gate driver U2 for Q3 (high side) / Q4 (low side), fault disable via PROT_DIS



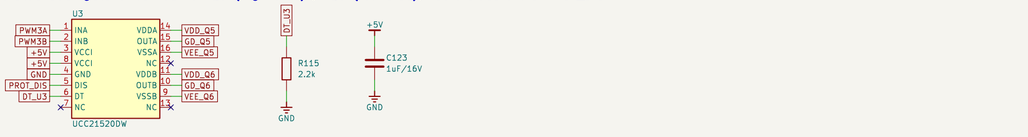

## Q3 isolated bias (+15 V / −3 V) and gate network



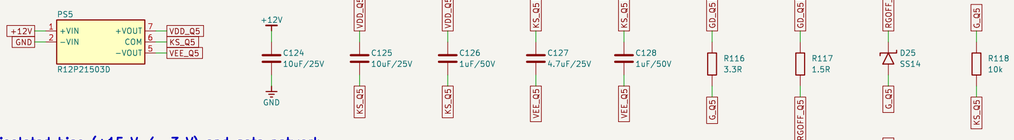

## Q4 isolated bias (+15 V / −3 V) and gate network



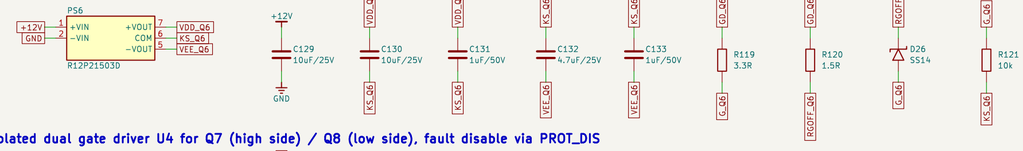

## Isolated dual gate driver U3 for Q5 (high side) / Q6 (low side), fault disable via PROT_DIS



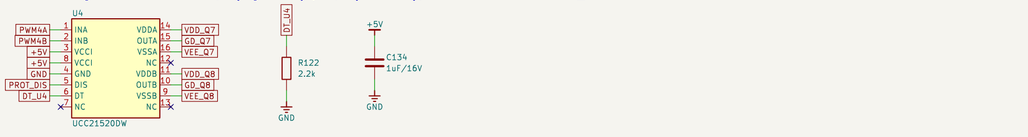

## Q5 isolated bias (+15 V / −3 V) and gate network



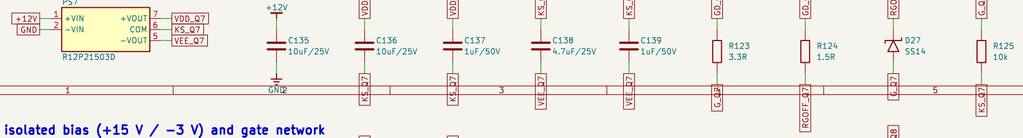

## Q6 isolated bias (+15 V / −3 V) and gate network




## Isolated dual gate driver U4 for Q7 (high side) / Q8 (low side), fault disable via PROT_DIS




## Q7 isolated bias (+15 V / −3 V) and gate network




## Q8 isolated bias (+15 V / −3 V) and gate network

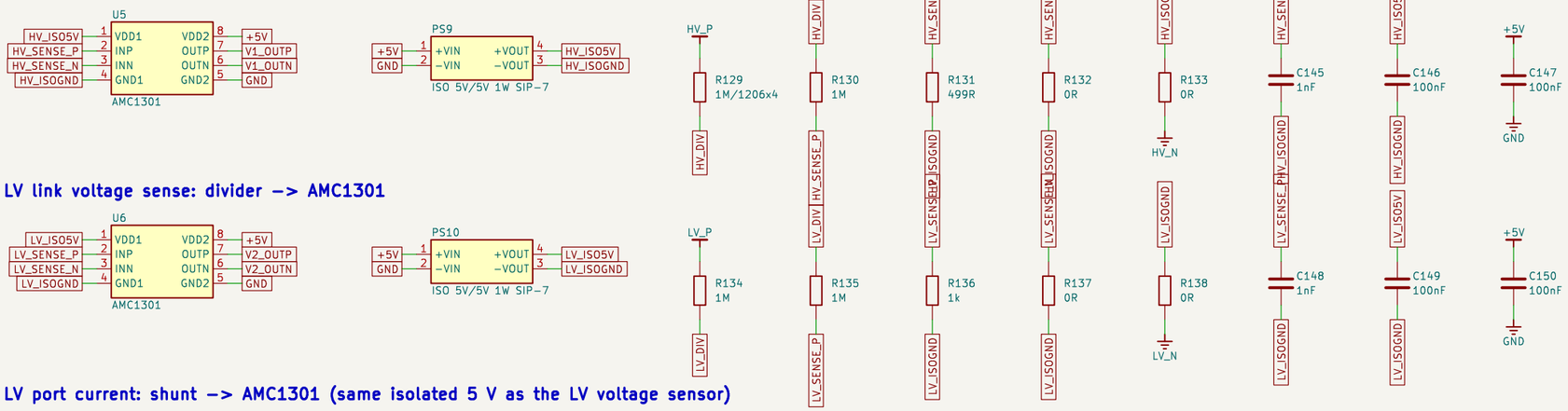
6.6 kW, 650–850 V <–> 300–450 V, 100 kHz, SiC DAB, SPS/EPS modulation
Gate Drive and Isolated Bias
**Open hardware — SiC PSFB reference design**

Sheet: /Gate Drive and Isolated Bias/
File: gate_drive.kicad_sch

***Title: SiC 6.6 kW Bidirectional Dual Active Bridge, 800 V / 400 V***

| Size: A2 | Date: 2026-10-09 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 4/6 |

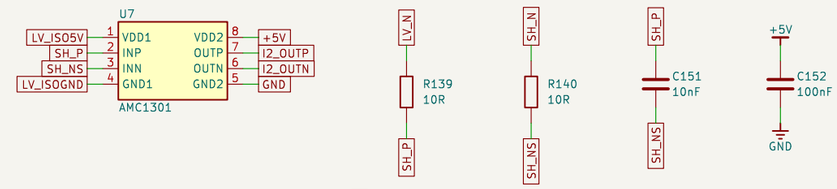
## HV link voltage sense: divider –> AMC1301 (isolated), isolated 5 V from a 1 W SIP module



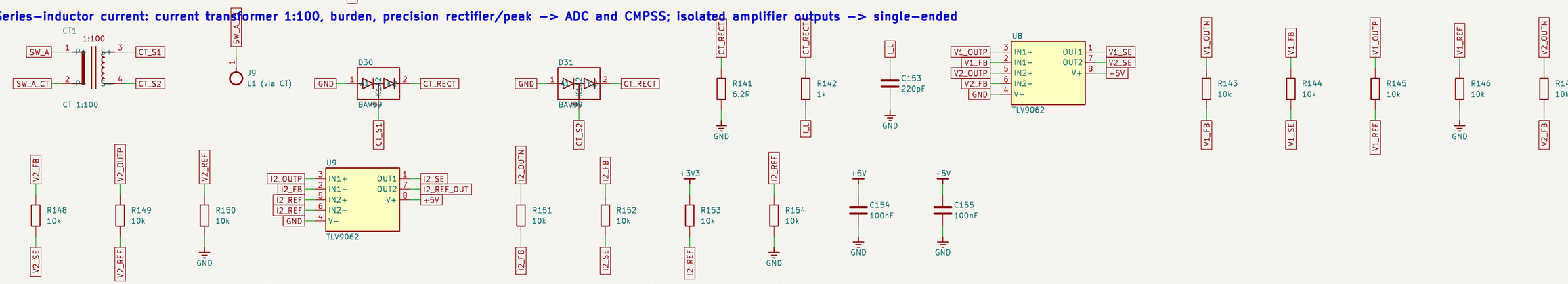

## LV link voltage sense: divider –> AMC1301



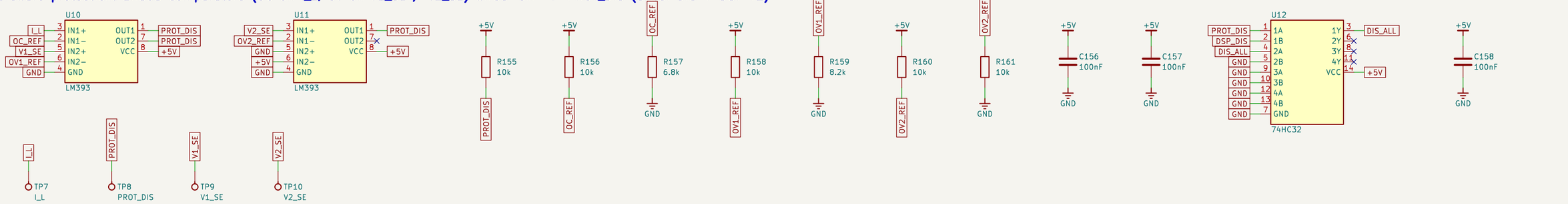

## LV port current: shunt –> AMC1301 (same isolated 5 V as the LV voltage sensor)




## Series–inductor current: current transformer 1:100, burden, precision rectifier/peak –> ADC and CMPSS; isolated amplifier outputs –> single–ended




## Hardware protection: LM393 comparators (OC on I_L, OV on V1_SE / V2_SE) wired–OR –> PROT_DIS (driver DIS + DSP TZ)



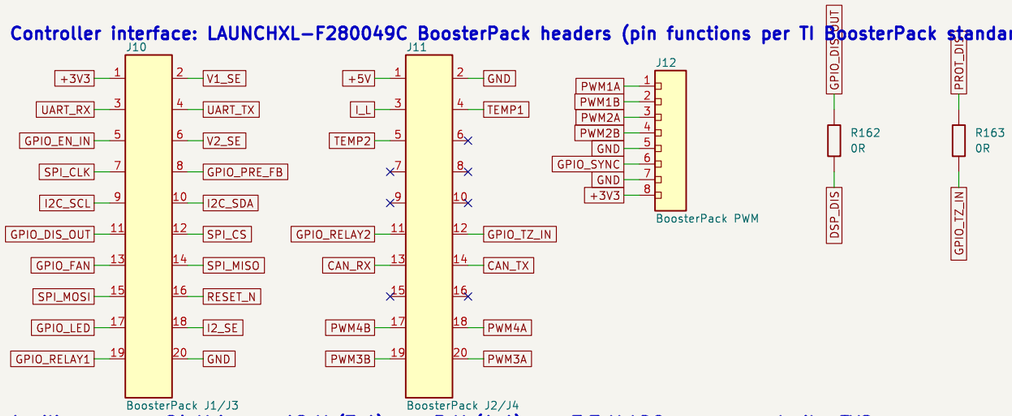

6.6 kW, 650–850 V <–> 300–450 V, 100 kHz, SiC DAB, SPS/EPS modulation
Sensing and Hardware Protection
**Open hardware – SiC PSFB reference design**

Sheet: /Sensing and Hardware Protection/
File: sensing_protection.kicad_sch

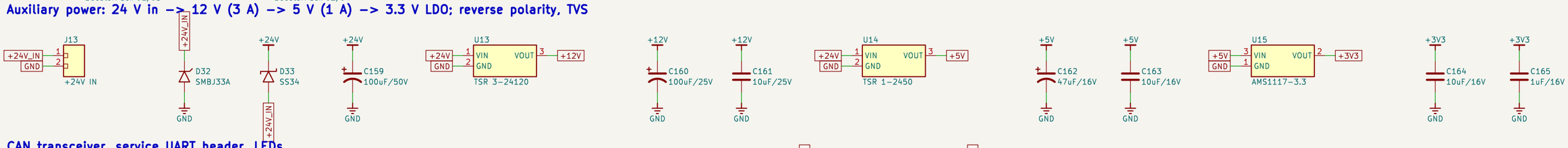
***Title: SiC 6.6 kW Bidirectional Dual Active Bridge, 800 V / 400 V***

| Size: A3 | Date: 2026–10–09 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4 | | Id: 5/6 |

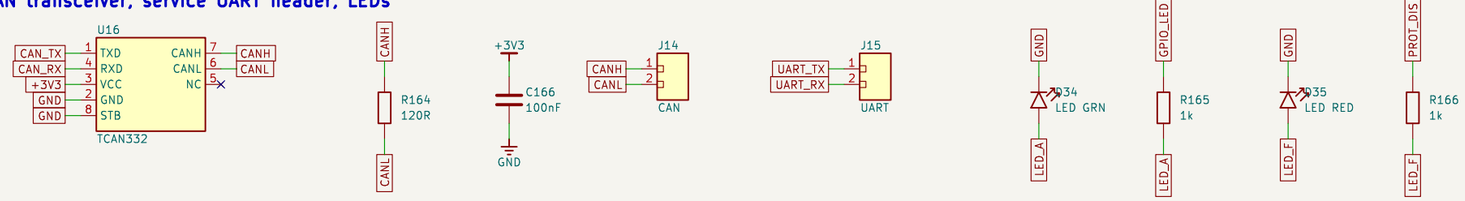
## Controller interface: LAUNCHXL-F280049C BoosterPack headers (pin functions per TI BoosterPack standard; F280049C GPIO mapping to be verified against SPRUII7)



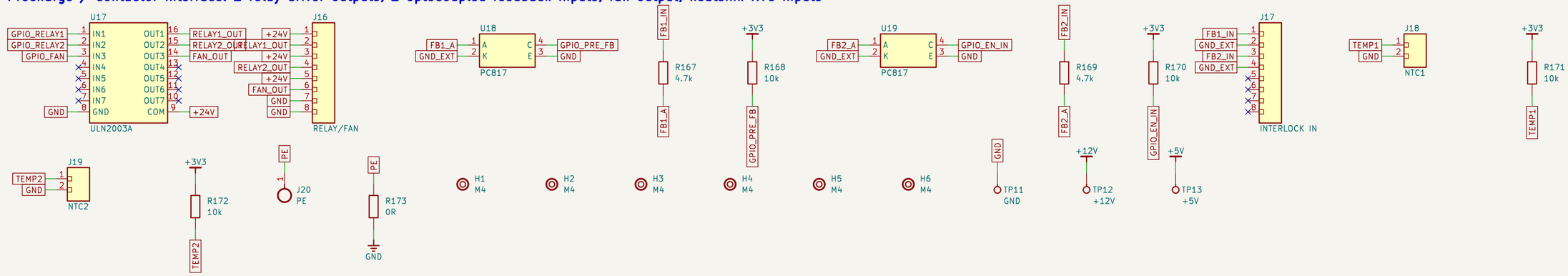

## Auxiliary power: 24 V in -> 12 V (3 A) -> 5 V (1 A) -> 3.3 V LDO; reverse polarity, TVS




## CAN transceiver, service UART header, LEDs




## Precharge / contactor interface: 2 relay driver outputs, 2 optocoupled feedback inputs, fan output; heatsink NTC inputs




6.6 kW, 650–850 V <–> 300–450 V, 100 kHz, SiC DAB, SPS/EPS modulation
Controller Interface, Aux Power, Comms
**Open hardware — SiC PSFB reference design**

Sheet: /Controller Interface, Aux Power, Comms/
File: control_aux.kicad_sch

***Title: SiC 6.6 kW Bidirectional Dual Active Bridge, 800 V / 400 V***

| Size: A3 | Date: 2026-10-09 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 6/6 |